International Journal of Food Science +Technology

International Journal of Food Science and Technology 2023, **58**, 3121–3134

Original article

Rapid recognition of processed milk type using electrical impedance spectroscopy and machine learning

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(Received 23 December 2022; Accepted in revised form 28 March 2023)

Summary Unscrupulous merchants would sell cheap, low-nutrition formula milk powder as pasteurised milk, or use it as a raw material in dairy products, for the purpose of making a profit. Currently, biochemical methods are utilised to identify the type of processed milk, which could involve chemical reagents, sample preparation and costly instruments. This paper investigates the utility of electrical impedance spectroscopy (EIS) in distinguishing different types of processed milk. Ultra-high temperature sterilised milk, pasteurised milk and formula milk powder were studied. The one-factor analysis of variance revealed that milk type had a significant influence on electrical properties, whereas sensory analysis, pH and total soluble solids showed poor discrimination of milk type. Recognition models were developed using Support Vector Classification, Partial Least Squares-Discriminant Analysis (PLS-DA), Gaussian Naive Bayes and Random Forest (RF). The nested cross-validation and the external validation set (N = 45) indicated the RF model had the highest predicting accuracy of 97.8%; whereas PLS-DA had a misclassification rate of 33.3% on positive samples and was, therefore, unsuitable for the authentication. The results demonstrated that the EIS data was featured enough for a small sample modelling (N = 75), and the established method was simple, fast and effective for liquid food classification.

Keywords Electrical impedance spectroscopy, Gaussian Naive Bayes, Partial Least Squares-Discriminant Analysis, processed milk type, Random Forest, Support Vector Classification.

Introduction

Milk is widely known for its valuable nutritional content, such as protein, fat, carbohydrate, sodium, vitamin and calcium (Collard & McCormick, 2021). As a result, milk has become an essential part of modern diets due to its capability to sustain human health and its low cost. In 2019, the global milk output exceeded 852 million tons per year (Bórawski *et al.*, 2021). To maintain the quality and meet the transportation requirements, there are three types of processed milk being produced: ultra-high temperature sterilised (UHT) milk, pasteurised milk and formula milk powder. Among the three, pasteurised milk is believed to be the most nutritious (Cai *et al.*, 2022) since it can retain its bioactive components, such as immunoglobulin, lactoferrin, lactalbumin and lactoglobulin, which can promote human immune system (Donovan, 2016; Bakar *et al.*, 2021).

With the advancement of food manufacturing, an increasing number of food processors produce dairy products, with milk as the primary raw material (Varnam & Sutherland, 2001). Unfortunately, some fraudulent practices are still prevailing. Dishonest merchants advertise certain products as containing high quality pasteurised milk when in reality they are using other types of milk, sometimes even deteriorating milk. Even more serious incidents occur when diabetic and hypertensive patients consume formula milk powder without realising that it contains high amounts of sugar and sodium. Therefore, it is of importance to come up with an accurate, simple and quick method of milk detection to supervise food safety.

In the past, researchers have taken various approaches to detect the quality of milk, such as NMR spectroscopy to evaluate the impact of

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doi:10.1111/ijfs.16440
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hyperbaric storage on the chemical composition of pasteurised milk (Lemos *et al.*, 2022), chromatography and mass spectroscopy to differentiate organic and UHT milk (Samarra *et al.*, 2021), or fluorescence and Fourier transform infrared spectroscopy to identify the origins of milk (Fragkoulis *et al.*, 2021). Additionally, marker peptides (Dongfei *et al.*, 2022) and stable isotopes approaches (Baohui *et al.*, 2022) can be used to distinguish between UHT and reconstituted milk. These methods can be used to analyse the distinction between various milks due to the diversity of principles they are based on, such as physics, biology, chemistry and spectroscopy. However, their high costs or the burden of sample pre-processing and spectra processing limit their applicability.

Compared with mainstream analytical techniques, electrical impedance spectroscopy (EIS) has several advantages. In Table 1, we compare EIS with other analytical techniques from four perspectives. In terms of its cost and ease of operation, EIS has been used to investigate a wide range of materials (Brahma *et al.*, 2012; Linz *et al.*, 2020; Xiang *et al.*, 2020; Watanabe *et al.*, 2021). This technology involves the application of alternating current signals at various frequencies to the item being measured, allowing for the determination of several electrical properties, including impedance, conductivity, capacitance and dielectric constant. EIS is increasingly being applied to characterise liquid foods, with research showing it to be useful for quantifying adulteration in apple juice (Banach & Żywica, 2021), measuring the dielectric and electrical properties of coconut water (Shah *et al.*, 2015), detecting adulteration in milk (Worku *et al.*, 2020), examining the impact of sugar cane and saccharin on the electrical properties of cider manalagi (Widodo *et al.*, 2019) and quantitatively estimating the soda ash as an adulterant in aqueous sucrose solution (Das *et al.*, 2019).

Machine learning is a branch of artificial intelligence and computer science, which focuses on the use of data and algorithms to imitate the way that humans learn, gradually improving its accuracy (Jordan & Mitchell, 2015). Among various algorithms, the classification algorithm that automatically orders or categorises data into different sets of classes has been applied in various fields (Mohamed, 2017; Soofi & Awan, 2017; Maxwell *et al.*, 2018). EIS combined with machine learning algorithms are widely used in the field of liquid food especially dairy product to detect additives or nutrients and quantify their concentration through the establishment of regression models between the substance and the electrical properties (Żywica & Banach, 2015; Lopes *et al.*, 2018; Das *et al.*, 2019; Widodo *et al.*, 2019; Ghodinde *et al.*, 2022). Furthermore, its feasibility for constructing classification models is also demonstrated, such as for the ripening degree of avocado (Islam *et al.*, 2018), fermentation process of black tea (Zhu *et al.*, 2019) and biological cell damage during freezing and thawing (Abie *et al.*, 2021). Although there is research using EIS and Support Vector Machine to classify milk with different amounts of fats (Ciosek *et al.*, 2006), but the study does not use fresh data to stress test the model. There is no reliable small sample modelling of EIS data for determining the type of processed milk, which has practical significance. Therefore, we aimed to test classical algorithms such as Support Vector Classification (SVC), Partial Least Squares-Discriminant Analysis (PLS-DA), Gaussian Naive Bayes (GNB) and Random Forest (RF) to see if they can be used to create a reliable classification model with a small sample set of EIS data. In previous studies, nested cross-validation and stress test have not been utilised to evaluate the performance of the established models. We will use these two methods to ascertain the reliability of the

Table 1 Comparison of electrical impedance spectroscopy with mainstream analytical techniques

Method	Equipment	Price (CNY)	Spectral processing	Sample pre-processing	Experienced operators
Electrical impedance spectroscopy	LCR Meter	1000 to 100 000	×	×	×
Fourier transform infrared spectroscopy	Fourier Transform Infrared Spectrometer	100 000 to 1 000 000	√	×	√
Nuclear magnetic resonance spectroscopy	Nuclear Magnetic Resonance Spectrometer	1 000 000 to 10 000 000	√	√	√
Mass spectroscopy	Mass Spectrometer	100 000 to 1 000 000	√	√	√
Raman spectroscopy	Raman Spectrometer	200 000 to 1 000 000	√	×	√

The exchange rate between CNY and US dollar is around 7:1, '√' represents that this condition is needed, while '×' represents that this condition is not needed.

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milk classification model, thus enabling food regulators to identify milk types quickly and reliably.

Materials and methods

Samples preparation

Seventy-five milk samples (twenty-five ultra-high temperature sterilised milk, twenty-five pasteurised milk and twenty-five formula milk powder) from seventy-five different brands were purchased from a local supermarket and stored at 8 °C prior to experimentation. These seventy-five brands covered the top-selling mainstream milk brands in the Chinese market, such as Yili, Mengniu, Bright, Sanyuan, Changfu, Anchor, Devondale, Weidendorf, etc.

Sensory analysis

A panel of 10 healthy individuals trained in milk evaluation was recruited. The panellists' requirements, as outlined in the literature (Cai *et al.*, 2022), were as follows: (i) five males and five females; (ii) aged 20–25; (iii) good health and no lactose intolerance; (iv) possess adequate language and textual expression abilities to accurately and properly convey evaluation results; (v) abstain from using strong cosmetics or perfume that would affect taste score; (vi) no smoking within 1 h prior to evaluation; (vii) abstain from eating hot pot, garlic, onion or other strongly flavoured foods within 1 h prior to evaluation.

Twenty-five pasteurised milk samples, twenty-five UHT milk samples and twenty five milk powder samples were analysed by panellists. Before tasting, each sample was brought to a temperature close to 20 °C after being exposed to room temperature for a while. Samples were placed in disposable paper cups without the label and the panellists rinsed with warm, clean water before each taste. The panellists used descriptive sensory analysis to score the sample (Murray *et al.*, 2001). Each sample was given a score of up to 10 points for texture, flavour, mouthfeel, colour, smell and personal preferences; the maximum combined score was 60 points.

Electrical impedance spectroscopy measurements

An EIS measurement system was employed to measure the electrical properties of a milk sample. The system consisted of an LCR meter (TH2816A, Tonghui Electronic Co., Ltd, Changzhou, China), two copper electrodes, a plastic container and a water bath box (ZSBB-712, Zhicheng, China), as depicted in Fig. 1. The liquid sample of milk powder was comprised of 25 g milk powder blended with 150 mL distilled water. A plastic container was filled with 40 mL of the sample, then it was placed in the water bath box, which was maintained at a stable temperature of 35 °C. The temperature needs to be kept constant throughout the experiment, because different temperatures affect the value of electrical properties and cause errors (Henningsson *et al.*, 2005). The series capacitance (Cs), parallel capacitance (Cp), impedance (Z), reactance (X), conductance (G), susceptance (B), dissipation factor (D), resistance (R) and phase angle (θ) of the milk sample were measured at twenty five different frequencies between 50 Hz and 20 kHz. The LCR meter had previously been calibrated and preheated for 30 min. Additionally, the total soluble solids (TSS) content and pH value of the samples were determined with a

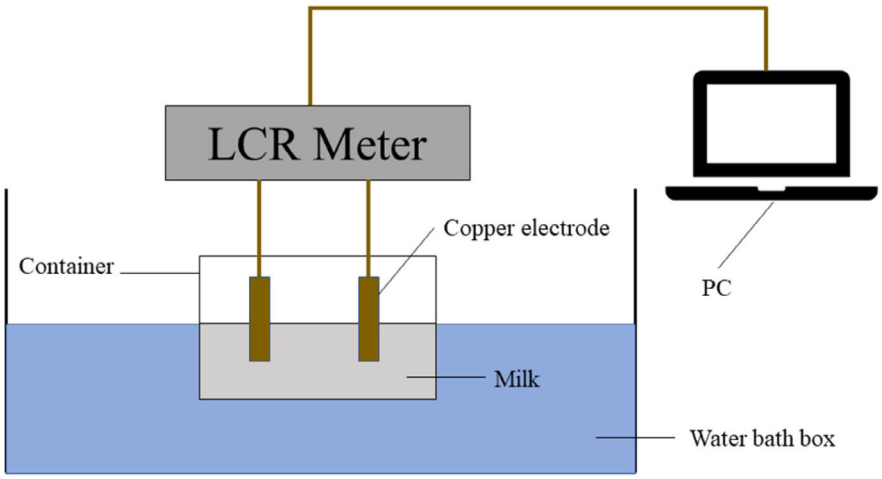

Figure 1 Schematic diagram of the electrical property measurement system.

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saccharometer (0 ~ 90°Brix, LICHEN, China) and pH meter (Five Easy Plus; METTLER TOLEDO, Zurich, Switzerland).

Statistical analysis

Probability–probability plots (Mushkudiani & Einmahl, 2007) and one-factor analysis of variance (ANVOA; Goldberg & Scheiner, 2020) were performed using the SPSS (International Business Machines Corporation, Armonk, NY, USA) software to test for normality and compare the mean values of the data from different groups, respectively.

Algorithms for processed milk type recognition

Principal component analysis

Principal component analysis (PCA) is used to summarise the information contained in large datasets into fewer, simpler to display and understand 'summary indices' (Abdi & Williams, 2010). Because this work attempts to model using small sample set, the number of predictors is greater than the number of samples (there are a large number of electrical properties at different frequencies). Therefore, the dimension of data set needs to be reduced by PCA.

Partial Least Squares-Discriminant Analysis

Partial Least Squares-Discriminant Analysis (PLS-DA; Lee *et al*., 2018) is a PLS regression method that is useful to identify subtle group differences when analysing multivariate datasets, particularly from spectra-based metabolomics. Similar to PCA, PLS-DA can also achieve dimension reduction. The difference is that PLS-DA is a supervised discriminant analysis method, which can better resolve the issues encountered in unsupervised analysis methods (PCA). When the difference between groups is large, and the difference within groups is small, unsupervised analysis can clearly differentiate between them; however, if the difference between groups is unclear but the difference within groups is large, it can be difficult for unsupervised analysis to distinguish them.

Support Vector Classification

Support vector machine (SVM) shows many unique advantages in solving small sample size, nonlinear and high-dimensional pattern recognition and is extended to other machine learning problems such as face recognition, pedestrian detection and text classification. Support Vector Classification (SVC; Yongli, 2012) is based on SVM for classifying linear and nonlinear datasets. After selecting a kernel function, it maps the data to a high-dimensional feature space and builds the best classification hyperplane in this space that separates the data that was poorly classified in the original plane. Linear, Polynomial, Gaussian and Sigmoid are the most commonly used kernel functions.

Gaussian Naive Bayes

Gaussian Naive Bayes (GNB; Zengjun *et al*., 2019) is a machine learning classification technique based on probability and Gaussian distribution. It is particularly useful for data with a continuous valued feature and a normal distribution. Each predictor is presumed to have an independent capacity to forecast output variables in GNB model. The ultimate forecast is the result of combining all the predictors.

Random Forest

A Decision Tree (Kingsford & Salzberg, 2008) is a non-parametric supervised learning algorithm utilised for both classification and regression tasks. To improve the effectiveness of Decision Trees, Random Forest (Biau & Scornet, 2016) was introduced. It involves the combination of multiple decision trees to form a 'forest', whereby they are fed with random features from the provided data set. Instead of relying on a single Decision Tree, the Random Forest takes predictions from all the trees and then votes on the best outcome.

Algorithm parameters

We used the eight electrical parameters (Cp, Cs, Z, θ, R, X, G and B) at six frequencies (60, 80, 5200, 6700, 10 000 and 14 000 Hz) as the predictors to construct the models, giving the data set a total of forty eight dimensions. The code and detailed parameter settings of each algorithm are placed in Supplementary Information. K-fold cross-validation ($K = 5$) (Fushiki, 2011) was applied to select the best model of the algorithms.

Validation

The nested cross-validation (Zhong *et al*., 2023) and external forty-five samples as stress test were applied to evaluate the goodness-of-fit and performance of the selected model, the following indicators were calculated: accuracy $\left(\text{accuracy} = \frac{\text{TP+TN}}{\text{TP+TN+FP+FN}}\right)$, sensitivity $\left(\text{sensitivity} = \frac{\text{TP}}{\text{TP+FN}}\right)$, specificity $\left(\text{specificity} = \frac{\text{TN}}{\text{FP+TN}}\right)$, precision $\left(\text{precision} = \frac{\text{TP}}{\text{TP+FP}}\right)$, recall $\left(\text{recall} = \frac{\text{TP}}{\text{TP+FN}}\right)$, F1 score $\left(\text{F1 score} = \frac{2(\text{Recall}\times\text{Precision})}{\text{Recall+Precision}}\right)$ where TP is true positive, TN is true negative, FP is false positive, FN is false negative (Trevethan, 2017), and the Root Mean Square Error (RMSE) (Nicolson & Paliwal, 2019)

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$\left(\mathrm{RMSE} = \sqrt{\frac{1}{M}\sum_{i=1}^{m}(y_i - \hat{y}_i)^2}\right)$, y_i is the actual value for observation i; $\hat{y}_i$ the predicted value for observation i; $\overline{y}$ the average value of the dependent variable, m is the total sample size. RMSE is used to measure the standard deviation of residuals. Sensitivity, specificity, recall, precision and *F1* score were to measure the reliability of classifiers when making negative and positive judgements.

Results and discussion

Nutrition database

We applied a nutrition database to analyse the nutritional content of 162 milk brands in order to understand and compare the quality of three types of processed milk available in the Chinese market. The results are displayed in Table 2. The data revealed that formula milk powder had higher carbohydrate, salt and calcium contents per mass compared to pasteurised and ultra-high temperature sterilised milk, but had the lowest protein content and cost out of the three. This implied that formula milk powder might contain additives such as sugar, calcium and sodium to save costs, enhance flavour and increase specific nutrition levels. ANOVA analysis in Table 2 showed that there were considerable variations ($P < 0.05$) in the nutritional content of the three types of milk, along with alterations in the electrical properties depending on the nutritional components. The result indicated that EIS might be able to recognise the type of processed milk.

Sensory analysis

Table 2 demonstrated substantial disparities in both price and nutrition among the three types of processed milk. To prevent dishonest merchants from substituting pasteurised or UHT milk with milk powder, which is cheaper and has lower nutritional value, food regulators need precise methods of distinguishing them. Thus, sensory analysis is a option. Figure 2 displays the scores that resulted from our evaluation of milk samples on the criteria of colour, texture, flavour, mouthfeel and preference. Pasteurised milk performed the best in six aspects, due to the fact that pasteurisation could cause minimal damage to milk composition and flavour (Bezie, 2019), and pasteurisation milk also had a recent production date. It was noticeable that milk powder was lower in terms of mouthfeel and preference compared to UHT milk and pasteurised milk. The main reason for this was that formula milk powder contained various additives that were used to increase specific nutrition or reduce manufacturing costs. This consequently led to a significant difference in taste compared to the other two kinds of milk. Furthermore, this caused least preference among panellists. It was achievable to distinguish pasteurised milk and formula milk powder by sensory analysis from our results, but pasteurised milk and ultra-high temperature milk only had difference in mouthfeel, flavour and preference, and the other three aspects were almost identical. It is difficult for the average consumer to differentiate between pasteurised milk and ultra-high temperature milk. Moreover, conducting a sensory analysis requires a professional sensory analysis laboratory and trained sensory analysts, making it hard for food regulators to detect milk products any time or place. Thus, EIS, being quick, simple, accurate and affordable, is preferred in this case.

Electrical impedance spectroscopy analysis

We chose the electrical properties at 80 Hz and 14 kHz to gauge the differences between means of the three processed milk. Probability–probability plot revealed that the data followed a normal distribution, permitting the application of ANOVA. According to the ANOVA results, there were considerable variations between the three types of milk samples, Cs ($P = 0.000$), X ($P = 0.000$), D ($P = 0.000$), Z ($P = 0.000$), θ ($P = 0.000$), G ($P = 0.000$) and B

Table 2 ANOVA on nutritional components of three types of processed milk

Nutrition and price	Pasteurised milk (N = 68)	UHT milk (N = 64)	Formula milk powder (N = 30)	P value
Energy (kJ/100 mL)	273.515 ± 22.6383	261.625 ± 44.0640	278.970 ± 19.8041	0.028
Protein (g)	3.334 ± 0.2237	3.364 ± 0.2547	2.983 ± 0.4189	0.000
Fat (g)	3.631 ± 0.5106	3.280 ± 1.1360	3.157 ± 0.8659	0.017
Carbohydrate (g)	4.854 ± 0.3150	4.883 ± 0.2265	6.477 ± 1.3702	0.000
Sodium (g)	0.05585 ± 0.009139	0.05669 ± 0.009456	0.06973 ± 0.012734	0.000
Calcium (g)	0.110118 ± 0.0097146	0.116734 ± 0.0119007	0.125438 ± 0.0222583	0.000
Price (CNY mL^{-1})	0.033125 ± 0.0186609	0.019181 ± 0.0091857	0.012705 ± 0.0031711	0.000

Mean ± SD, $P < 0.05$ indicates significant difference.

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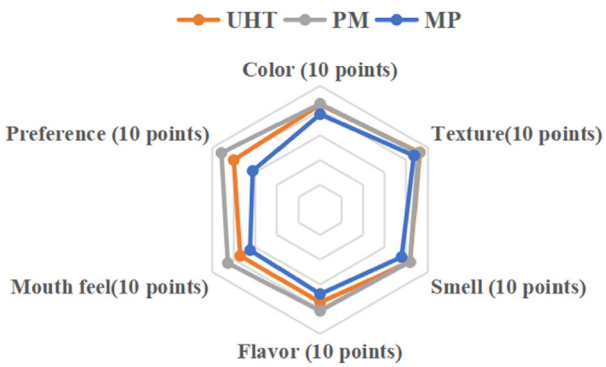

Figure 2 Sensory analysis of ultra-high temperature sterilised milk (UHT), pasteurised milk (PM) and formula milk powder (MP).

(P = 0.000). However, in a *post hoc* multiple comparison (Aslam & Albassam, 2020), the disparities in Z, G and B were mainly because of discrepancies between milk powder and the other two types of milk. Thus, these parameters were not helpful in distinguishing pasteurised milk and UHT milk. To establish this distinction, we used additional techniques such as pH and TSS detection. ANOVA results revealed that there were no significant differences in the pH values of pasteurised milk and UHT milk (P = 0.07 and P = 0.110, respectively). As such, comparing the pH value and TTS of milk rendered it difficult to determine the type of processed milk. We conducted frequency-dependent spectra of Cs, X and D in the range of 50 Hz to 20 kHz to visually display the differences between the three types of processed milk.

The frequency dependence of Cs is depicted in Fig. 3, showing a sharp decrease in Cs in the frequency range of 50–2000 Hz due to the higher alternating current resistance of the capacitance at higher frequencies. This trend gradually faded away after 2000 Hz due to the smooth flow of high frequency current in the sample. It could be seen that the Cs of milk powder was significantly higher than those of the other two types of milk. The dielectric constant, a parameter related to the degree of polarisation, can be calculated using the formula $C = \varepsilon S/4\pi k$, and its value is proportional to the value of Cs. Milk conductivity, which is affected by metal ions such as K+, Na+ and Ca+, is positively correlated with the liquid dielectric constant (Mabrook & Petty, 2003). Therefore, ions that can increase the polarity of the liquid may be added to milk powder, as shown in Table 2 of the nutrition database. This was why the milk powder had a distinctive Cs due to its higher conductivity than the other two types of milk. The varying sterilising processes and storage durations are the primary reasons for the differences between pasteurised milk and UHT milk. The longer storage period and higher sterilisation temperature of UHT milk are responsible for the deformation of fat globules and breaking down of the casein micelles (Mabrook & Petty, 2003), causing its higher Cs, as shown in Fig. 3.

Figure 4 manifests the connection of X (absolute value) as pasteurised milk > UHT milk > milk powder. The equation $X = 1/(2\pi fC)$ demonstrates that X is inversely proportional to C. The low-frequency portion (from 1000 to 5000 Hz) of Fig. 4 clearly displays this discrepancy. Therefore, the spectrum can be acquired with a narrow frequency range for practical milk quality assessment, which would lower the equipment purchase price, reduce the number of acquisition points and speed up the detection.

The frequency dependence of D is illustrated in Fig. 5. The value of D is proportional to Cs and Rs according to the equation. $D = 2\pi fCsRs$. Therefore, as depicted in Fig. 5, the distinguishment brought about by D among the three types of milk is more promising than that brought about by Cs alone. D is thus a more featured physical identifier for recognising the type of milk.

These results showed that it was feasible to differentiate between different milk types using electrical properties. Therefore, in order to build an accurate and intuitive processed milk type recognition model, we employed four classic machine learning algorithms to completely explore the differences among the multivariate data.

Machine learning

Support Vector Classification

Because we could not determine whether the data set was linearly separable, we chose a linear kernel and a nonlinear kernel: 'Linear' and 'Poly' for the SVC algorithm. Linear kernel function is defined as: $k < x_1, x_2 > == < x_1, x_2>$. The linear kernel function is mainly used in the case of linear separability, and the effect is ideal. When using SVM for classification, the linear kernel function is usually first tried to do classification. Poly kernel function is defined as: $k < x_1, x_2 > = (< x_1, x_2 > + 1)^d$ (d is the data dimension). Poly kernel function can map the input space of low-dimensional to the feature space of high-dimensional, but the disadvantage is that there are too many parameters and high computational complexity. We attempted to choose electrical properties at one or two frequencies in an effort to lower the number of eigenvalues. However, this resulted in a significant loss in model accuracy and demonstrated that the electrical properties at each frequency contained separate information. We, therefore, used the data that was processed using PCA (PC1 = 41.7%, PC2 = 33.1%, PC3 = 12.2%, these three PCs were used) (Tonglin & Baijian, 2016) into the SVC algorithm for modelling in

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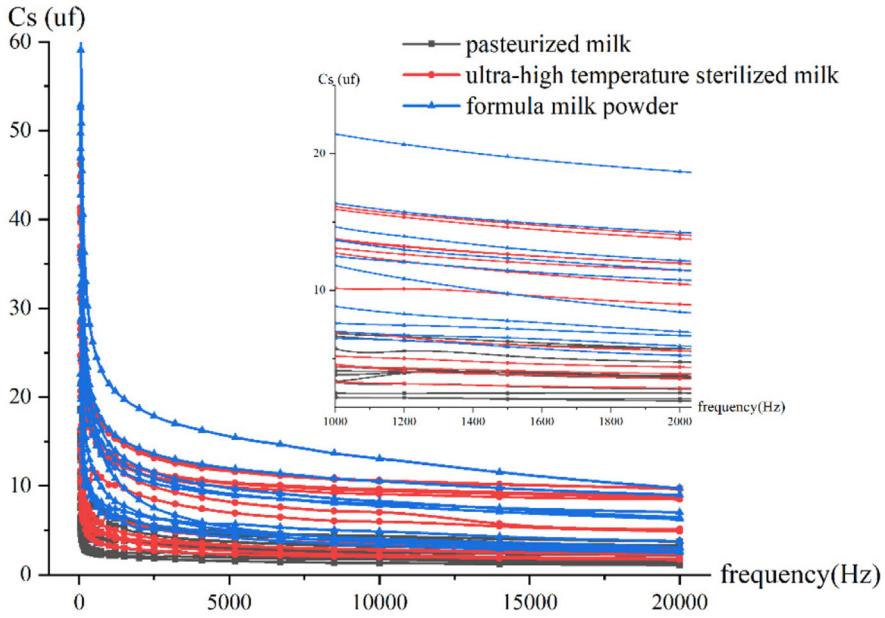

Figure 3 The frequency-dependent spectra of the measured series capacitance (Cs) of three types of processed milk at 35 °C.

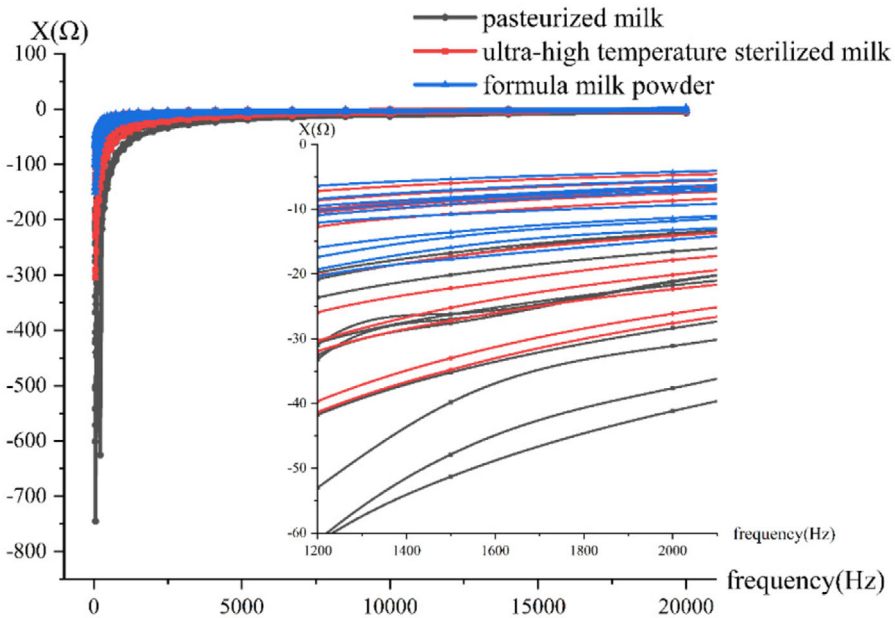

Figure 4 The frequency-dependent spectra of the measured reactance (X) of three types of processed milk at 35 °C.

order to reduce the error caused by noise information and improve the accuracy. The linear and poly Kernel functions achieved the same accuracy of 93.3% on the test set. In confusion matrices of SVC in Fig. S1, the columns represented the prediction results, and the rows represented the actual classification. So, the numbers on the diagonal indicated that the test result were correct. We could see that both the two kernel functions had only one misclassification on UHT milk.

In order to comprehensively evaluate and select the appropriate model, in addition to accuracy, sensitivity and specificity are also important indicators. We

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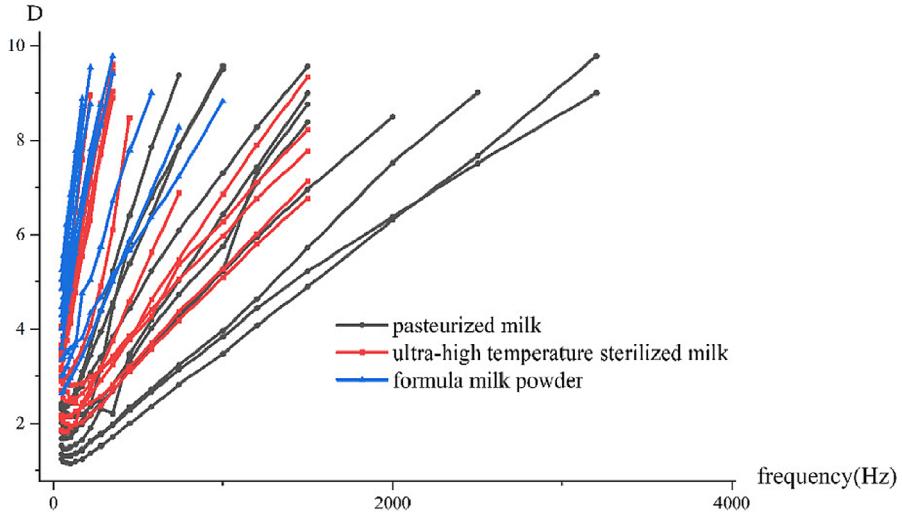

Figure 5 The frequency-dependent spectra of the measured dissipation factor (*D*) of three types of processed milk at 35 °C.

regarded UHT milk and formula milk powder as positive, pasteurised milk as negative, so sensitivity represented the classifier's ability to recognise UHT milk and formula milk powder, while specificity represented the classifier's ability to recognise pasteurised milk. According to the confusion matrices in Fig. S1, the sensitivity was calculated as 0.889 (Linear) and 0.909 (Poly), respectively, and the specificity was calculated as 1.00 in both.

Partial Least Squares-Discriminant Analysis

Similar to PCA, PLS-DA can also achieve dimension reduction. We have tested both the original data and data processed by PCA in the PLS-DA model. The results demonstrated that using data processed with PCA could lead to higher accuracy. Therefore, we fed the PCA-processed data into the PLS-DA model. The accuracy of the PLS-DA classification was 73.3%. Figure 6 illustrated the classification results. The pasteurised milk and milk powder samples were separated clearly into two distinct groups; however, the dimension reduced characteristics of some UHT milk samples were close to the ones of the milk powder, and others were close to the pasteurised milk, as shown in Fig. 6. The confusion matrix in Fig. S2 verified the results displayed in Fig. 6. According to the confusion matrix in Fig. S2, we could calculate that the sensitivity and specificity are both equal to 1.

Gaussian Naive Bayes

The accuracy of GNB classification was 86.6%. Fig. 7 depicts the GNB classification borders, wherein a region of the same colour corresponds to a region of the same category sample. From the results, we could

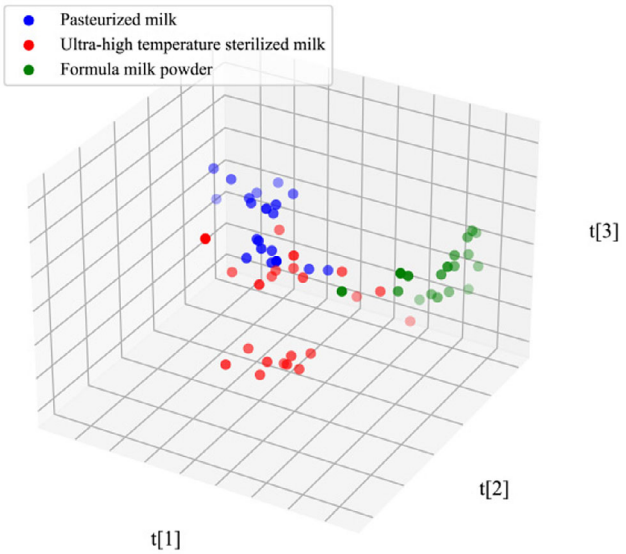

Figure 6 PLS-DA score plot.

see that the region of milk powder and pasteurised milk was far away, and the regions were separated by the region of UHT milk, indicating that the quality gap between milk powder and pasteurised milk was large, and the quality of UHT milk was in between. UHT milk samples were relatively concentrated in Fig. 7, indicating that UHT sterilisation and packaging could ensure the stability of milk quality. Because milk powder is added with different additives for each producer, pasteurised milk is easy to spoil and lead to unstable quality, so their corresponding points were

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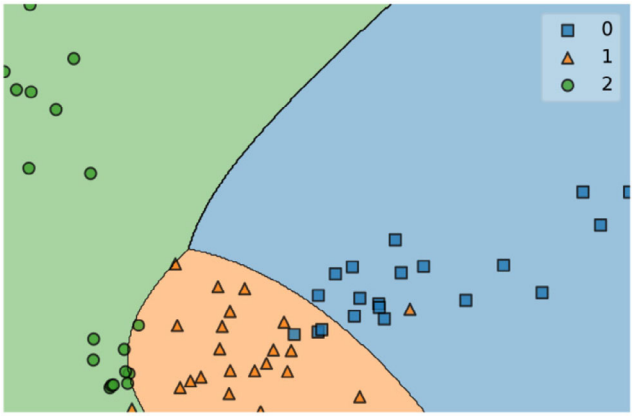

Figure 7 Boundaries of GNB classification (0 = pasteurised milk, 1 = ultra-high temperature sterilised milk, 2 = formula milk powder).

relatively dispersed in the figure. Figure S3 showed a confusion matrix of the test set. The sensitivity was calculated to be 0.900 and the specificity was 0.800.

Random forest

Random Forest is a machine learning algorithm based on Decision Trees, which uses the power of multiple decision trees to make decisions. The decision tree is based on the Gini index scoring method (Frees *et al.*, 2011). Its formula is as follows: $(\mathrm{Gini(p)} = \sum_{k=1}^{K} \mathrm{p}_k(1-p_k) = 1-\sum_{k=1}^{K} p_k^2)$. The smaller the 'Gini' value, the higher the purity, which means that the probability of a randomly selected sample being misclassified in the sample set is smaller. The classification result of one decision tree in the RF is illustrated in Fig. 8, the same colour tone represents the same group. The accuracy of RF model on the test set was 93.3%. According to the confusion matrix in Fig. S4, the sensitivity was calculated as 0.909, and the specificity was calculated as 1.00.

Validation results

The reliability of mathematical models constructed using small sample sets has been widely contested (Raudys & Jain, 1991; Jain & Zongker, 1997; Dai, 2008; Cohen *et al.*, 2015). As such, validating the small sample model is of importance. *K*-fold cross-validation is a common technique employed to test for overfitting; yet when this is done on small sample sets, it can easily lead to biased performance estimation (Braga-Neto & Dougherty, 2004; Isaksson *et al.*, 2008; Vabalas *et al.*, 2019). *K*-fold cross-validation is better suited for calibrating and selecting the best performing model rather than validating the model when using a small sample set (Hawkins *et al.*, 2003). To reduce the risk of overfitting, stress test and other methods that yield conservative measures of uncertainty should be applied.

Nested cross-validation has proven to provides unbiased performance estimates with small sample size (Vabalas *et al.*, 2019). It utilises an inner and outer loop to decrease the risk of model scoring bias. The inner loop is a cross-validation intended to locate the best model hyperparameters for fitting the outer loop. The outer loop furnishes the inner loop with training data and preserves a portion of data for testing the model inside the inner loop. The performance of different classification models on nested cross-validation is shown in Table 3. The sample data set utilised in this work included eight electrical properties at six different frequencies, belonging to three separate groups. This necessitated that the recognition model should be adept at handling relatively high-dimensional data variables. RF works through many classification trees, generating and aggregating results together; the diversity of the trees contributes to the predictive accuracy of the samples. In theory, as long as there are enough decision trees, a high accuracy can be achieved. As shown in Table 3, RF achieved the best accuracy, RMSE, demonstrating that EIS based RF model was capable of classifying processed milk types with a sample size lower than the total number of predictor variables, which was consistent with previous studies (Ham *et al.*, 2005; Díaz-Uriarte and Alvarez de Andrés, 2006; Luan *et al.*, 2020).

A new set of forty five samples (fifteen pasteurised milk, fifteen ultra-high temperature sterilised milk and fifteen milk powder samples) was employed as a stress test to assess the model's generalisation capability. Pre-processing was conducted using the same procedure outlined in the Materials and Methods section. No labels were assigned to the validation data. The results are shown in Table 4. When a true sample is incorrectly classified, the milk producer or seller are likely to doubt the detecting result and use alternative methods to ascertain the sample's class. However, they will not raise a concern if a false sample is identified as true. The precision, recall and F1 score, which calculated by the confusion matrix in Fig. 9, are to study such problems (Miao & Zhu, 2022). Pasteurised milk is most likely to be replaced by UHT milk and milk powder because of the high cost of pasteurised milk in comparison to those two products. As a result, we considered pasteurised milk to be negative and the other two types of milk to be positive. The higher the precision, the fewer negative samples are incorrectly classified. The higher the recall, the fewer positive samples are missed. F1 score is defined as the harmonic mean of the model's precision and recall, a high F1 score implies the high stability of the model. In our

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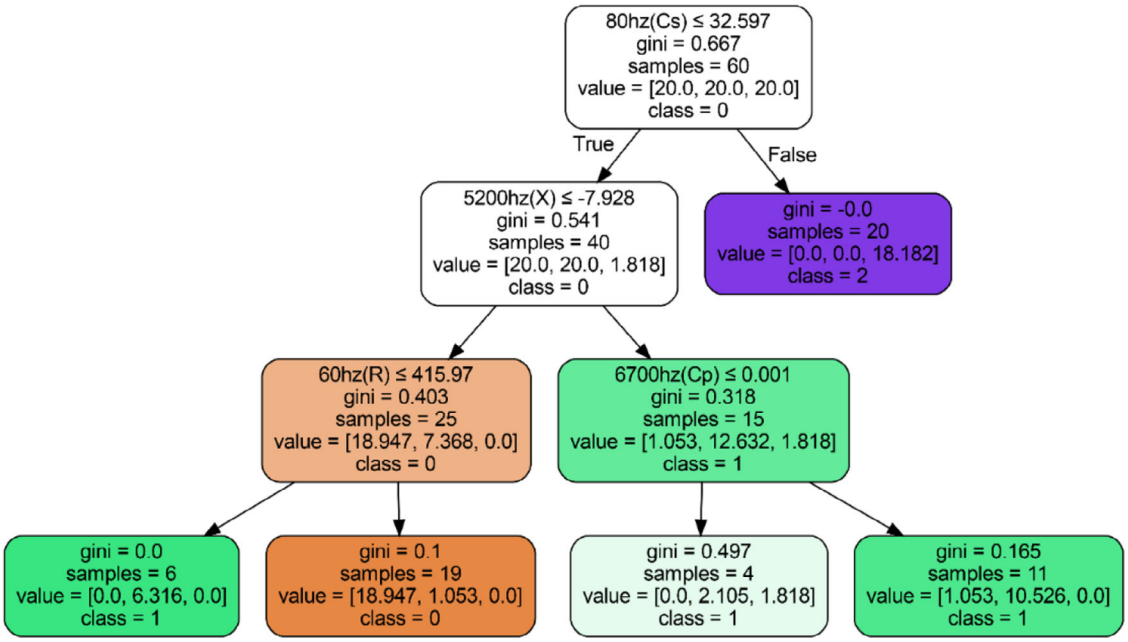

Figure 8 Classification result of a decision tree (0 = pasteurised milk, 1 = ultra-high temperature sterilised milk, 2 = formula milk powder).

Table 3 The performance of different classification models on nested cross-validation (N = 75)

Algorithms	Accuracy	RMSE
SVC (Linear)	0.880	0.3464
SVC (Poly)	0.853	0.3834
PLS-DA	0.760	0.4899
GNB	0.893	0.3271
RF	0.920	0.2828

Table 4 The performance of different classification models on the external validation (N = 45)

Algorithms	Accuracy	Precision	Recall	F1 score
SVC (Linear)	0.867	1.00	0.960	0.980
SVC (Poly)	0.889	1.00	0.926	0.962
PLS-DA	0.778	1.00	0.909	0.952
GNB	0.911	0.964	0.964	0.964
RF	0.978	1.00	0.967	0.983

opinion, milk powder or UHT milk being sold as pasteurised milk should be recognised as much as possible for consumer rights, so recall is important than precision in processed milk type detection. The confusion matrices in Fig. 9 showed that although some models had low accuracy such as PLA-DA and SVC, they still had high sensitivity for pasteurised milk when the errors were mainly concentrated on positive samples. Eventually, by comparing the accuracy, precision, recall and confusion matrices on external validation, the RF and GNB algorithm, which had the high recalls was suitable for adulteration problems; RF, PLS-DA and SVC were good at identifying pasteurised milk; RF had the highest accuracy and F1 score, allowing it to achieved recognition of different types of processed milk.

In 2018, a study was performed by Liu *et al.* (2018) to authenticate the organic milk utilising portable near-infrared spectroscopy (NIRS). To discriminate organic, pasture milk and conventional milk, they applied spectral process auto-scaling, mean-centring, smoothing, first derivative, log 10 transformation and multiple scattering correction, then used PLS-DA to construct the discrimination model with the sample size of 87. They found resolution of portable NIRS was lower than benchtop NIRS; however, it could achieve the same clarification as benchtop NIRS did. Comparing to NIRS, EIS is not able to obtain the nutrients profile of milk, yet LCR METER is simple to operate, does not need experienced personals and spectral process. In our study, we differentiated among three types of the milk with an accuracy of 97.8% (The accuracy using NIRS in their study was 95%) utilising EIS spectra at 5–20 KHz. The minimum price of LCR meter that can provide this frequency range is about $500, such a low price helps to promote the use

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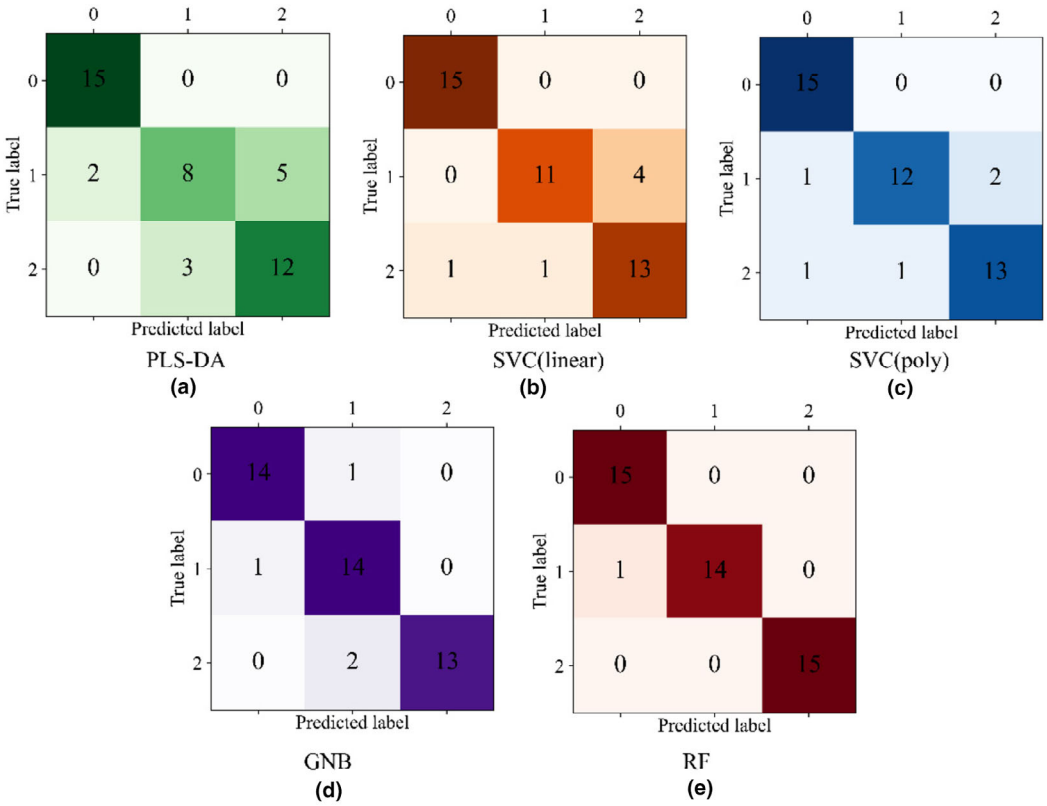

Figure 9 The confusion matrices of the external validation set (N = 45): (a) PLS-DA, (b) SVC (linear), (c) SVC (poly), (d) GNB and (e) RF. (0 = Pasteurised milk, 1 = Ultra-high temperature sterilised milk, 2 = Formula milk powder).

of this method in underdeveloped and developing countries. Moreover, their study used PLS-DA to construct the model, but PLS-DA has been proven to be an inappropriate method of authentication (Rodionova *et al.*, 2016). In case the new sample does not belong to any class of training datasets, PLS-DA is unable to properly define the membership of the sample. The PLS-DA result on the external validation in Table 4 also supported this point. Our study offers food regulators a low cost, simple and non-destructive alternative approach to distinguish milk categories.

According to previous studies, bacteria, additives and storage time all have an impact on the electrical properties of milk (Bacher *et al.*, 2012; Özcan *et al.*, 2020; Scandurra *et al.*, 2022), so the models we have built also have the potential to apply in these areas. Milk can become contaminated and spoiled during milking, transport, processing and storage, where a variety of bacteria may develop. The count of enteric bacillus, lactic acid bacteria and low-temperature bacteria are essential indicators of milk quality. To ensure the quality, we will create a quantitative regression model for these bacteria and electrical properties, as well as a recognition model for spoiled milk in our next research.

Conclusions

The electrical properties measurement combined with machine learning were developed and applied successfully on identification of processed milk types. The electrical properties not only showed significant differences between milk power and other two types of milk through ANOVA, but also provided enough prediction variables for the classification models. The performance of the SVC, PLS-DA, GNB and RF algorithms were compared in the study. According to the evaluation indexes of validations, the RF model offered superior classification and prediction accuracy on the three groups, while the PLS-DA was unsuitable for authentication purposes due to its misclassification rate of 33.3% on positive samples. Our work demonstrated that the EIS data was featured enough for a small sample set, and this established method was potential for daily inspection of liquid foods.

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Funding information

This work was supported by the Natural Science Foundation of Fujian Province of China (grant number 2022J01821); the Natural Science Foundation of China (grant numbers 32172339, 11705068).

Conflict of interest

The authors declare no competing interests.

Author contributions

Ziyu Huang: Data curation (equal); formal analysis (equal); writing – original draft (equal); writing – review and editing (equal). **Yanghao Xiao:** Data curation (equal); writing – original draft (equal); writing – review and editing (equal). **Yuhui Xiao:** Data curation (equal); formal analysis (equal); validation (equal); writing – original draft (equal); writing – review and editing (equal). **Honghao Cai:** Conceptualization (lead); data curation (equal); formal analysis (equal); funding acquisition (lead); investigation (equal); methodology (equal); project administration (lead); writing – original draft (equal); writing – review and editing (equal). **Hui Ni:** Resources (equal); supervision (equal).

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Peer review

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Data availability statement

Some or all data, models or code that support the findings of this study are available from the corresponding author.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Appendix S1. Supplementary information on algorithm codes and the confusion matrix on test set.

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